Half Wave Rectifier

Name: Mohammad Omar Salameh – 2020240078

Mgs030@live.aul.edu.lb

Course: CCE214 Electronics I

Instructor: Dr. Engr. Khaled Mouchref

Academic Year: 2026-2027

Fall 2026

Table of Contents

1. Introduction 3
3. Required Components and Equipment 3
4. Basic Circuit 3
5. Working Principle 4
5.1 Positive Half-Cycle 4
5.2 Negative Half-Cycle 4
6. Main Calculations 6
6.1 Average DC Output Voltage 6
6.2 RMS Output Voltage 6
6.3 Peak Load Current 6
6.4 Peak Inverse Voltage 6
10. Advantages and Limitations 6
Advantages 6
Limitations 7
11. Applications 7
13. Conclusion 7

1. Introduction

A half-wave rectifier is a basic electronic circuit that converts an alternating-current (AC) signal into a one directional, pulsating output. The circuit uses a diode as a one-way path for current.

During one half of the AC cycle, the diode conducts and current reaches the load. During the opposite half-cycle, the diode blocks current. This project demonstrates the operation of a PN-junction diode, forward and reverse biasing, rectification, ripple, and simple capacitor filtering.

3. Required Components and Equipment

Item	Suggested Value / Type	Purpose
AC source	10 V peak, 50 Hz function generator	Provides the sinusoidal input
Diode	1N4007	Performs rectification
Load resistor	1 kΩ	Represents the load
Oscilloscope	Two-channel	Displays input and output waveforms
Digital multimeter	DC/AC measurement	Measures voltage and current

4. Basic Circuit

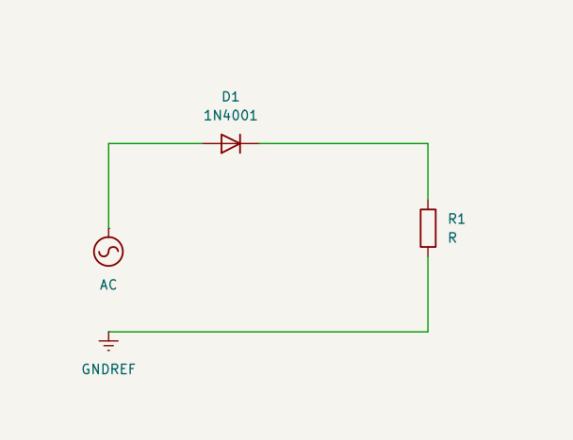


Figure 1 Basic half-wave rectifier circuit.

The diode is connected in series with the load resistor. The output voltage is measured across the load. The orientation shown passes the positive half-cycle of the input signal. Reversing the diode would pass the negative half-cycle instead.

5. Working Principle

5.1 Positive Half-Cycle

When the input voltage becomes positive, the anode of the diode is at a higher potential than the cathode. The diode becomes forward biased and begins to conduct. Current flows through the diode and then through the load resistor. For a silicon diode such as the 1N4007, the forward voltage drop is approximately 0.7 V under ordinary laboratory conditions.

$$V_{out} \approx V_{in} - V_D$$

$$For\ a\ silicon\ diode:\ Vout \approx Vin - 0.7\,V$$

For example, if the input peak voltage is 10 V:

$$V_{out(peak)} \approx 10 - 0.7 = 9.3\,V$$

5.2 Negative Half-Cycle

When the AC input changes polarity, the diode becomes reverse biased. In this condition it blocks current, apart from a very small leakage current. Because current through the load is almost zero, the output voltage is approximately zero during this half of the cycle.

$$I \approx 0\ A \quad \text{and} \quad Vout \approx 0\ V$$

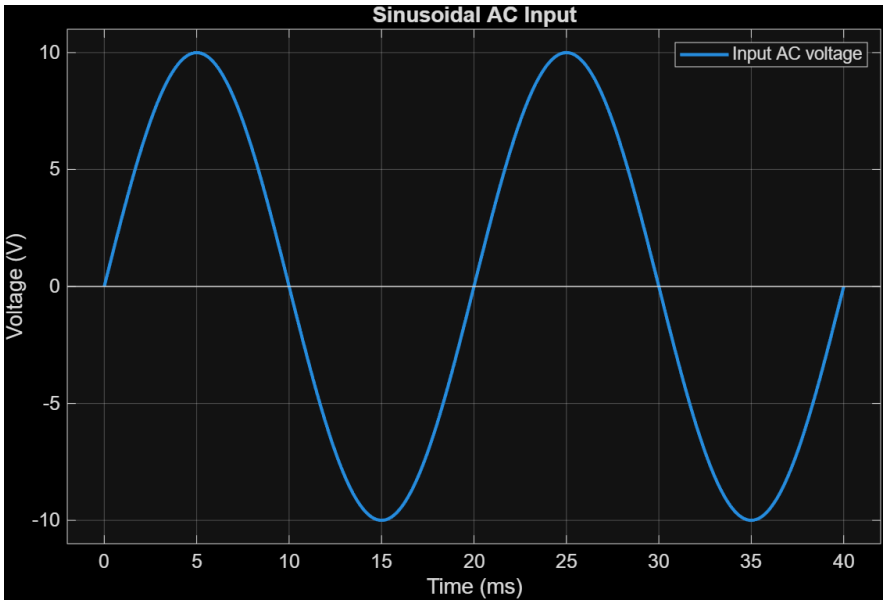


Figure 2 Sinusoidal AC input voltage.

The equation used is:

$$v_{in}(t) = 10sin(2\pi(50)t)$$

So the waveform has:

- **Peak voltage is 10 V**
- **Peak-to-peak voltage 20V**

- **Frequency:** 50 Hz
- **Period:** 20ms

At $t = 5, ms$, the voltage reaches approximately **+10 V**, at 15 ms it reaches **−10 V**, and the complete waveform repeats every **20 ms**.

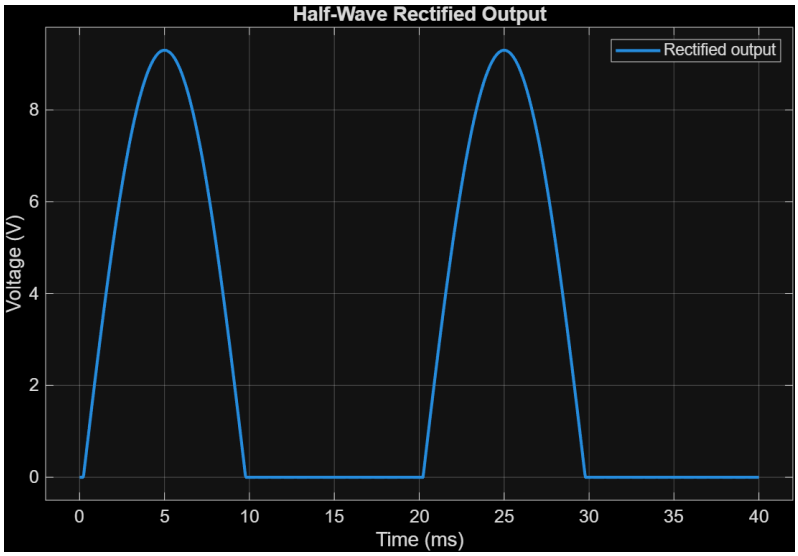


Figure 3 Half-wave rectified output. The negative half-cycles are blocked

This models the diode behavior. During the **positive half-cycle**, the diode conducts:

$$V_{out} = V_{in} - 0.7$$

During the **negative half-cycle**, the diode blocks current:

$$V_{out} = 0$$

So with a **10 V peak input**, the output peak is approximately:

$$V_{out}\,(pk) = 10 - 0.7 = 9.3\,V$$

6. Main Calculations

For an ideal half-wave rectifier with a sinusoidal input having a peak value of Vm:

6.1 Average DC Output Voltage

$$V_{DC} = \frac{V_m}{\pi} \approx 0.318\, V_m$$

Assuming that $V_m = 10\, V$:

$$V_{DC} = \frac{10}{\pi} \approx 3.18\, V$$

6.2 RMS Output Voltage

$$V_{RMS} = \frac{V_m}{2}$$

For $V_m = 10\, V$

$$V_{RMS} = \frac{10}{2} = 5\, V$$

6.3 Peak Load Current

$$I_m = \frac{V_m}{R}$$

For $Vm = 10\, V\ and\ R = 1\, k\Omega = 1000\, \Omega$, the ideal peak current is:

$$I_m = \frac{10}{1000} = 0.010\, A = 10\, mA$$

Including an approximate 0.7 V diode drop:

$$I_{peak} \approx \frac{10 - 0.7}{1000} = 9.3\, mA$$

6.4 Peak Inverse Voltage

Peak inverse voltage (PIV) is the maximum reverse voltage the diode must withstand while it is not conducting. For a simple half-wave rectifier without a filter, the required PIV is approximately equal to the peak input voltage.

$$PIV \approx V_m$$

10. Advantages and Limitations

Advantages

- Very simple circuit with few components.
- Low cost and easy to build on a breadboard.
- Useful for demonstrating diode behavior and basic rectification.

- Suitable for low-power and signal-processing demonstrations.

Limitations

- Only one half of the AC waveform is used.
- The unfiltered output contains significant ripple.
- Its theoretical maximum rectification efficiency is lower than that of a full-wave rectifier.
- A large filter capacitor may be needed when a smoother DC output is required.

11. Applications

- Simple low-power DC supplies.
- Signal detection and demodulation circuits.
- Wave-shaping and peak-detection demonstrations.
- Educational laboratory experiments for studying diodes and rectifiers.
- Circuits where high efficiency is not the main requirement.

13. Conclusion

The half-wave rectifier shows how a single PN-junction diode can convert an AC waveform into a one-directional output. The diode conducts during the forward-biased half-cycle and blocks current during the reverse-biased half-cycle. As a result, the load receives only one half of the original sinusoidal signal. Adding a capacitor across the load stores energy near the waveform peak and releases it while the diode is off, which reduces ripple and produces a smoother DC voltage. The experiment provides a direct connection between diode theory, waveform analysis, practical measurements, and the first stage of a basic DC power supply.